Electric Flux Density

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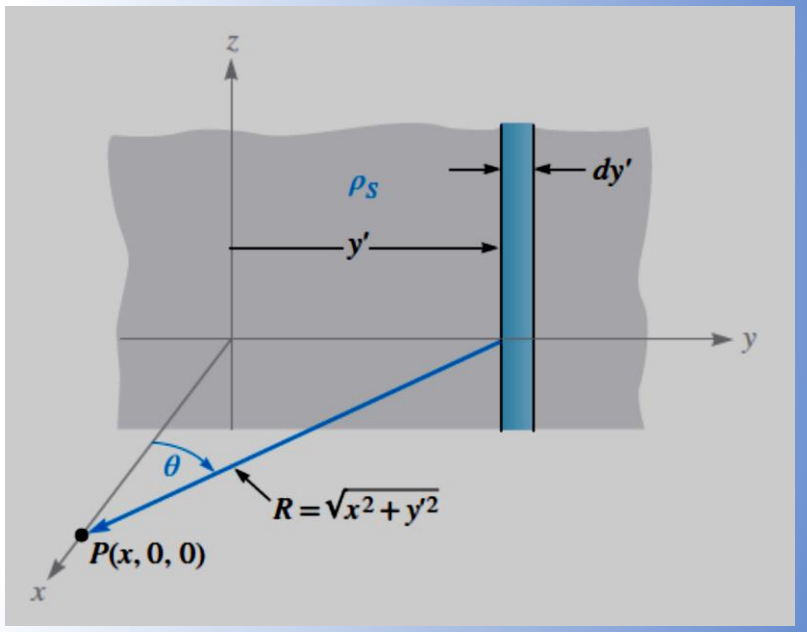
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Field of a Sheet of Charge

- Another basic charge configuration is the infinite sheet of charge having a uniform density of ρ_S C/m2.
- The charge-distribution family is now complete: point (Q), line (ρ_L), surface (ρ_S), and volume (ρ_v).
- Let us examine a sheet of charge, which is placed in the yz plane.



- The plane can be seen to be assembled from an infinite number of line charge, extending along the z axis, from $-\infty$ to $+\infty$.

Field of a Sheet of Charge

$$\mathbf{E} = \frac{\rho_s}{2\varepsilon_0}\mathbf{a}_N$$

- A unit vector a_N, which is normal to the sheet and directed away from it.
- The field of a sheet of charge is constant in magnitude and direction.
- It is not a function of distance.
- Fact: The electric field is always directed away from the positive charge, into the negative charge.

2.25 Find **E** at the origin if the following charge distributions are present in free space: point charge, 12 nC, at $P(2, 0, 6)$; uniform line charge density, 3 nC/m, at $x = -2$, $y = 3$; uniform surface charge density, 0.2 nC/m^2, at $x = 2$.

$$\mathbf{E} = \left[\frac{(12 \times 10^{-9})}{4\pi\epsilon_0}\frac{(-2\mathbf{a}_x - 6\mathbf{a}_z)}{(4+36)^{1.5}}\right] + \left[\frac{(3 \times 10^{-9})}{2\pi\epsilon_0}\frac{(2\mathbf{a}_x - 3\mathbf{a}_y)}{(4+9)}\right] - \left[\frac{(0.2 \times 10^{-9})\mathbf{a}_x}{2\epsilon_0}\right]$$

$$= \underline{-3.9\mathbf{a}_x - 12.4\mathbf{a}_y - 2.5\mathbf{a}_z \text{ V/m}}$$

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\mathbf{a}_R$$

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho$$

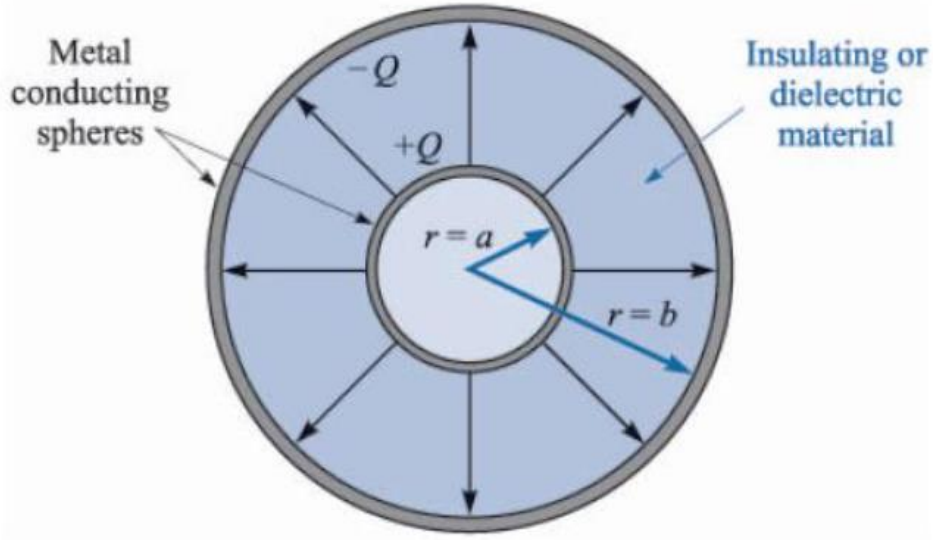
$$\mathbf{E} = \frac{\rho_s}{2\varepsilon_0}\mathbf{a}_N$$

Electric Flux Density

- About 1837 Michael Faraday, was interested in static electric fields and the effect of various insulating materials on these fields
- This led to his famous invention, the electric motor
- He found that if he moved a magnet through a loop of wire, an electric current flowed in the wire.
- The current also flowed if the loop was moved over a stationary magnet
- Changing magnetic field produces an electric field

Electric Flux Density

- In one of his experiments, Faraday had a pair of concentric metallic spheres constructed, the outer one consisting of two hemispheres that could be firmly clamped together
- He also prepared shells of insulating material (or dielectric material), which would occupy the entire volume between the concentric spheres





Electric Flux Density

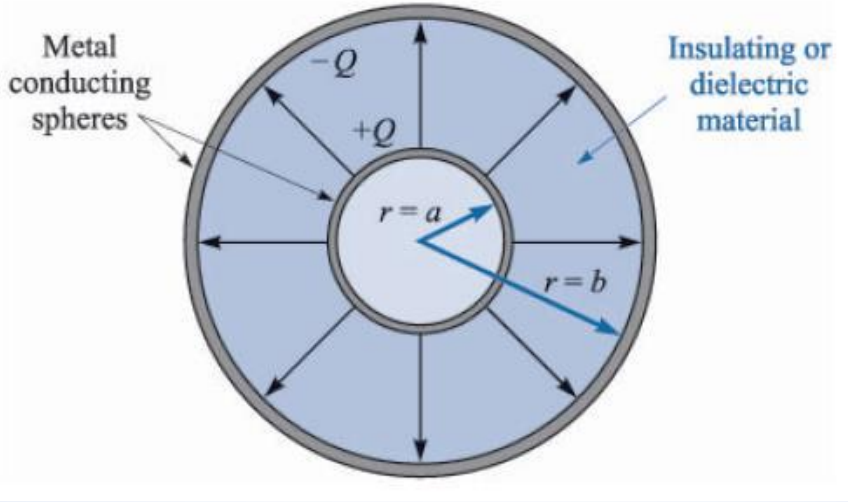
- Faraday found out, that there was a sort of “charge displacement” from the inner sphere to the outer sphere, which was independent of the medium.
- We refer to this flow as displacement, displacement flux, or simply electric flux.

$$\psi = Q$$

- where ψ is the electric flux, measured in coulombs, and Q is the total charge on the inner sphere, also in coulombs.

Electric Flux Density

- At the surface of the inner sphere, ψ coulombs of electric flux are produced by the given charge Q coulombs, and distributed uniformly over a surface having an area of $4\pi a^2$ m^2.
- The density of the flux at this surface is $\psi/ 4\pi a^2$ or $Q/ 4\pi a^2$ C/m2.



Electric Flux Density

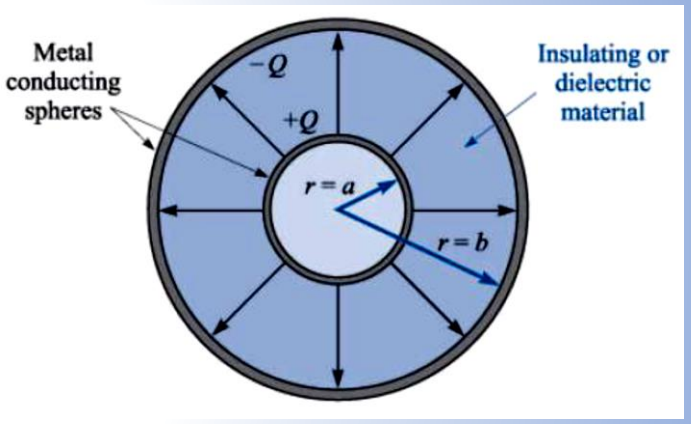
- The new quantity, electric flux density, is measured in C/m^2 and denoted with D.
- The direction of D at a point is the direction of the flux lines at that point.
- The magnitude of D is given by the number of flux lines crossing a surface normal to the lines divided by the surface area.

Electric Flux Density

- Referring again to the concentric spheres, the electric flux density is in the radial direction :

$$\mathbf{D}\big|_{r=a} = \frac{Q}{4\pi a^2}\mathbf{a}_r \qquad \text{(inner sphere)}$$

$$\mathbf{D}\big|_{r=b} = \frac{Q}{4\pi b^2}\mathbf{a}_r \qquad \text{(outer sphere)}$$



- At a distance *r*, where $a \le r \le b$,

$$\mathbf{D} = \frac{Q}{4\pi r^2}\mathbf{a}_r$$

- If we make the inner sphere smaller and smaller, it becomes a point charge while still retaining a charge of Q. The electric flux density at a point r meters away is still given by:

$$\mathbf{D} = \frac{Q}{4\pi r^2}\mathbf{a}_r$$

Electric Flux Density

- The radial electric field intensity of a point charge in free space is (as seen previously):

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\mathbf{a}_r$$

- Therefore, in free space, the following relation applies:

$$\mathbf{D} = \varepsilon_0 \mathbf{E}$$

- For a general volume charge distribution in free space:

$$\mathbf{E} = \int_{\text{vol}} \frac{\rho_v dv}{4\pi\varepsilon_0 R^2}\mathbf{a}_R$$

$$\therefore \quad \mathbf{D} = \int_{\text{vol}} \frac{\rho_v dv}{4\pi R^2}\mathbf{a}_R$$

Electric Flux Density

- Example
  Find the electric flux density at a point having a distance 3 m from a uniform line charge of 8 nC/m lying along the *z* axis in free space.

$$\mathbf{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\mathbf{a}_\rho \Rightarrow \mathbf{D} = \frac{\rho_L}{2\pi\rho}\mathbf{a}_\rho = \frac{8\times10^{-9}}{2\pi\rho}\mathbf{a}_\rho = \frac{1.273\times10^{-9}}{\rho}\mathbf{a}_\rho \text{ C/m}^2$$

For the value ρ = 3 m,

$$\mathbf{D} = \frac{1.273\times10^{-9}}{3} = 4.244\times10^{-10}\mathbf{a}_\rho \text{ C/m}^2 = \underline{0.424\mathbf{a}_\rho \text{ nC/m}^2}$$

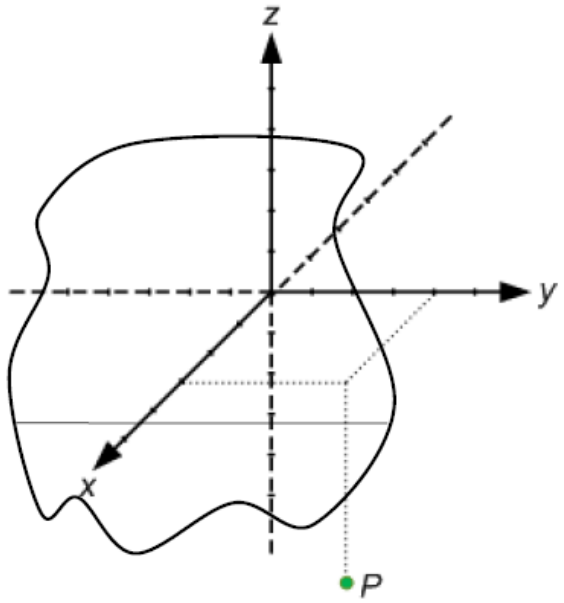
Electric Flux Density

- Example
  Calculate D at point $P(6,8,-10)$ produced by a uniform surface charge density with $\rho_s = 57.2\ \mu C/m^2$ on the plane $x = 9$.

$$\mathbf{E} = \frac{\rho_s}{2\varepsilon_0}\mathbf{a}_N \quad \Rightarrow \mathbf{D} = \frac{\rho_s}{2}\mathbf{a}_N = \frac{57.2\times10^{-6}}{2}\mathbf{a}_N = 28.6\mathbf{a}_N\ \mu C/m^2$$

At $P(6,8,-10)$,

$$\mathbf{a}_N = -\mathbf{a}_x \quad \Rightarrow \mathbf{D} = \underline{\underline{-28.6\mathbf{a}_x\ \mu C/m^2}}$$



Practice problem

D3.2. Calculate **D** in rectangular coordinates at point $P(2, -3, 6)$ produced by: (a) a point charge $Q_A = 55$ mC at $Q(-2, 3, -6)$; (b) a uniform line charge $\rho_{LB} = 20$ mC/m on the x axis; (c) a uniform surface charge density $\rho_{SC} = 120\ \mu C/m^2$ on the plane $z = -5$ m.

Ans. $(a)\ 6.38\mathbf{a}_x - 9.57\mathbf{a}_y + 19.14\mathbf{a}_z\ \mu C/m^2$; $(b)\ -212\mathbf{a}_y + 424\mathbf{a}_z\ \mu C/m^2$; $(c)\ 60\mathbf{a}_z\ \mu C/m^2$


Thank
you